



Aris Machina

assembling atoms with bits, using AI.

Introduction & Product

September 2026

assembling **atoms** is a **complex** problem

manufacturing
industrialization
R&D

energy
robotics
defense
bio-pharma
semiconductors

Aris Machina

Founded 2025 in Stockholm

Before – Apple, Palantir,
Northvolt, Tesla, Alpine-BWT
Rolls Royce, Pepsi, Spotify

Team – Physics, Machine Engineering,
Distributed Software, Machine Learning,
Materials Science, Manufacturing Operations

Customers – Advanced Automotive, Energy,
Defense, Robotics, Semiconductors,
Bio-pharma & MedTech, Automotive etc.

R&D Intelligence

*AI that accelerates
discovery.*



Industrialization Ontology

*From design to production,
nothing is lost.*



Shopfloor Intelligence

*AI systems never
sleep.*

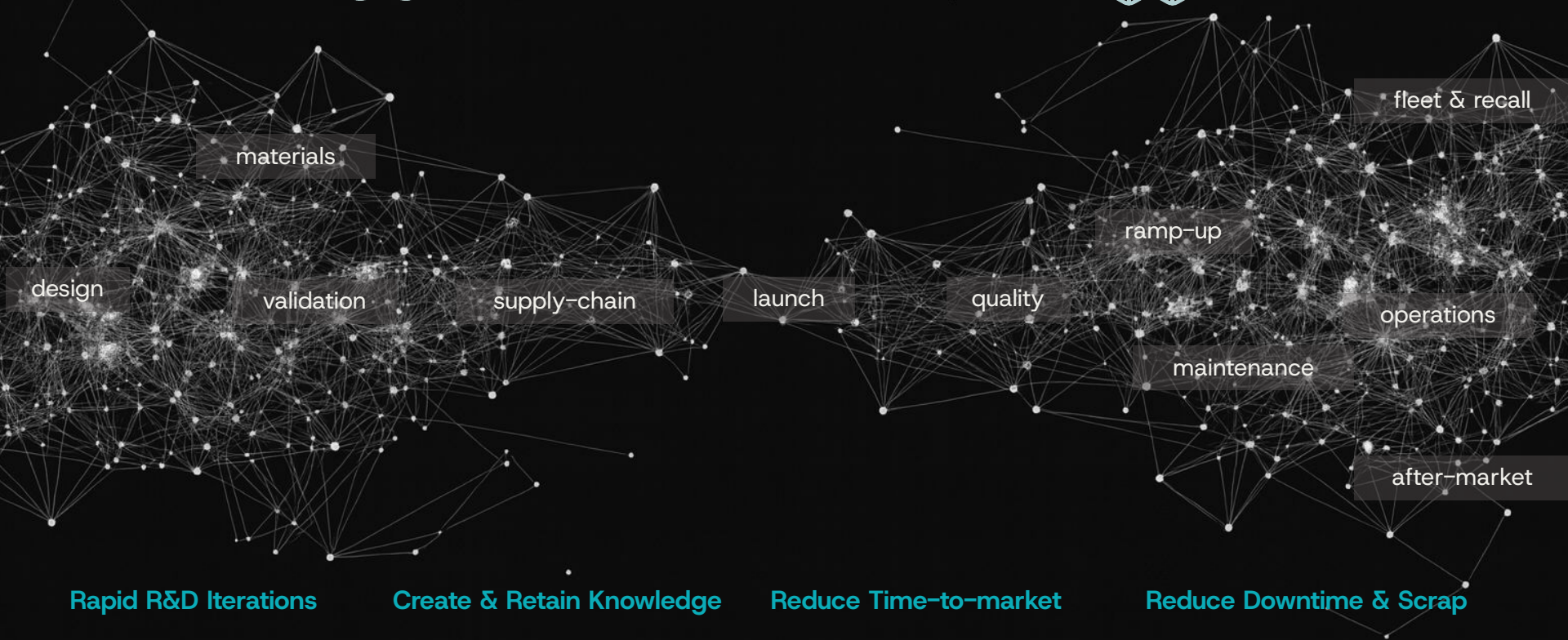


Industrial Super-Intelligence

enabling concept-to-production lifecycle for frontier industrials

Protos 

 **Gemba**



The Operating Platform for Building Physical things

Automotive Mfg.

Advanced materials

Batteries & Cells

Semiconductors

Aerospace

Biopharma

 Aris Machina

for Design

 **Protos**

FRONTIER R&D TEAMS · PROCESS DESIGN

Design, simulate and optimize physical systems from first principles.

Mapping Physics in 2 hours instead of 30 days

THE ENGINE
Physics models

×

Live data

↓

Agentic OS

 Aris Machina

for Operations

 **Gemba**

SHOPFLOOR TEAMS · OPERATIONS

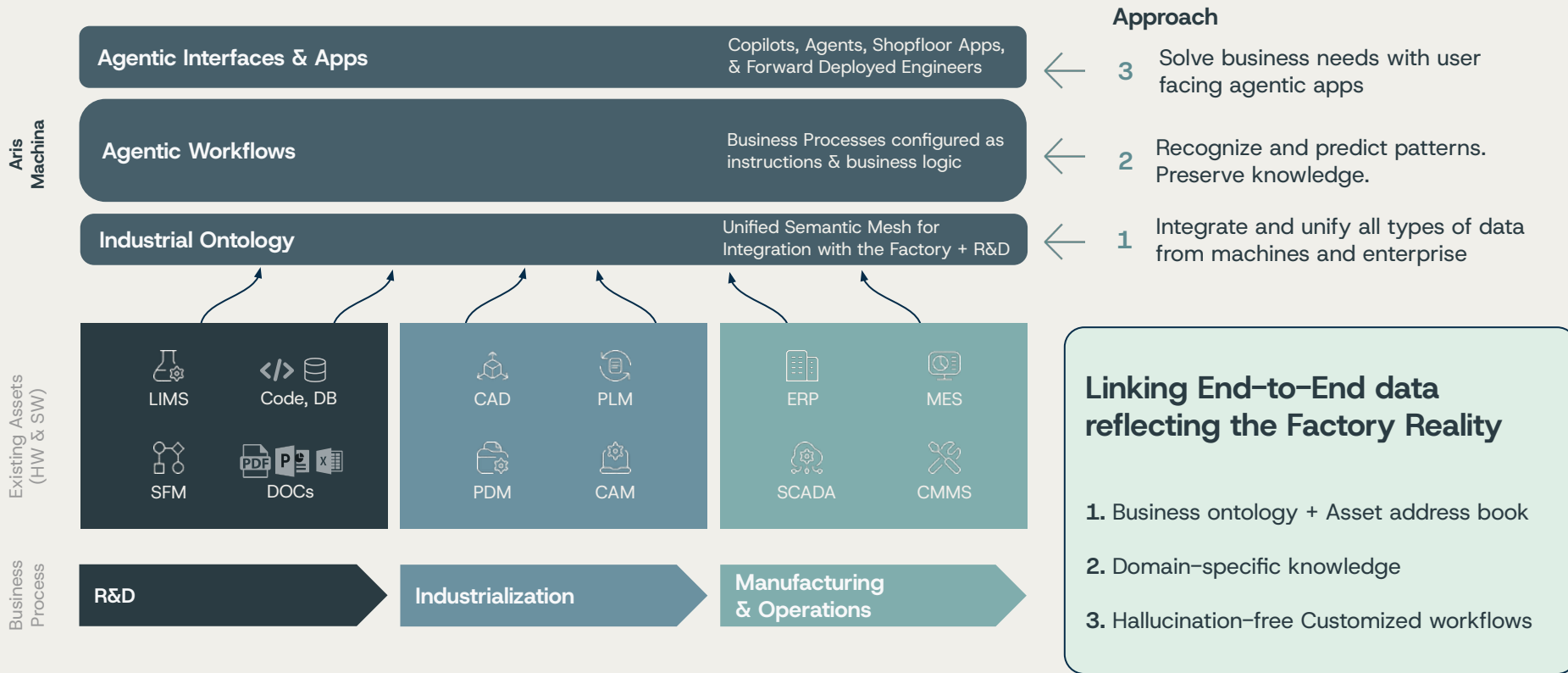
Real-time troubleshooting and process control on live machines.

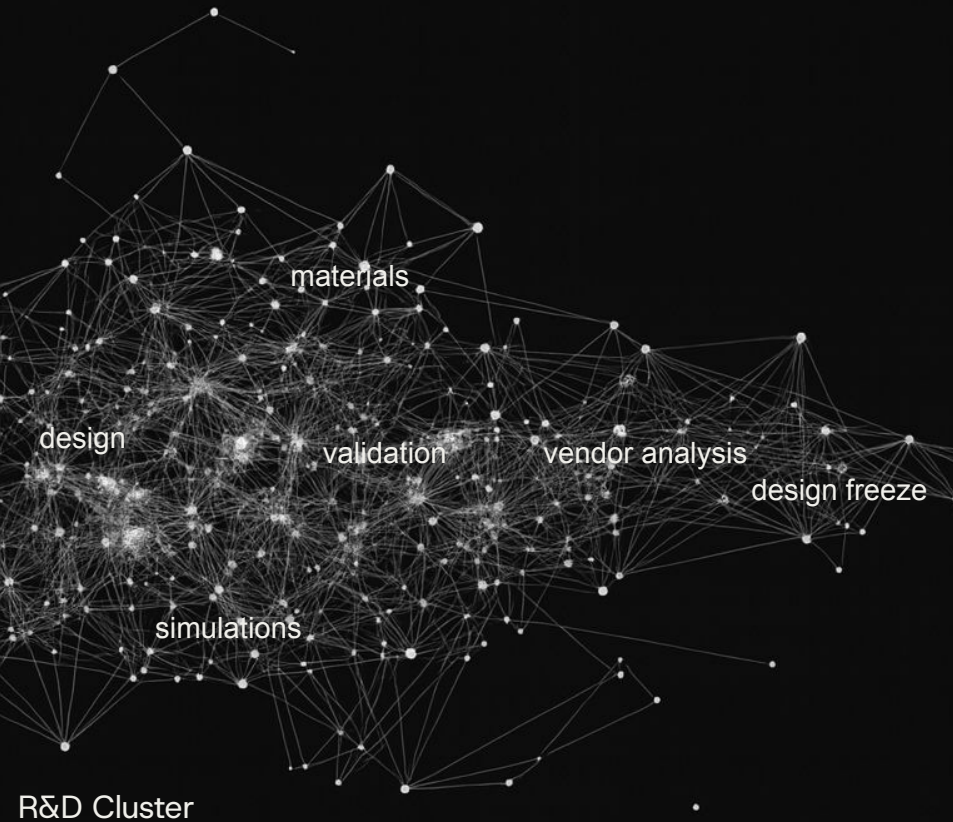
20+ Downtime predictors in live in 1 week

Core Principle — physics fused with live data — pointed at the **shopfloor** and at the **lab**.

Aris Platform

Enabling a unified and holistic **human workforce** to deliver end-to-end product lifecycles





Frontier Industry R&D has Long Development Cycles

Computational tools are monolithic and stuck in the past.



Your AI Co-Engineer for Industrial R&D

5-10x Faster design freeze cycles

7x Faster Knowledge Retrieval



The AI-native Development Environment for Industry

**Built for industrial-grade R&D,
only bounded by imagination.**

Industrial-grade Master Data Management

Design, simulate, validate, and document in one environment.
Full lifecycle – data, simulations, tests in a unified context.

Enterprise-ready & AI-native

collaborative, connector-ready, access controlled, private tenant deployment, and compliance-grade data handling.

Built for the Industrial Science Community

From solo researcher to expert consultants to large R&D teams —
Explorer, Pro, Team, or Enterprise.

minutes

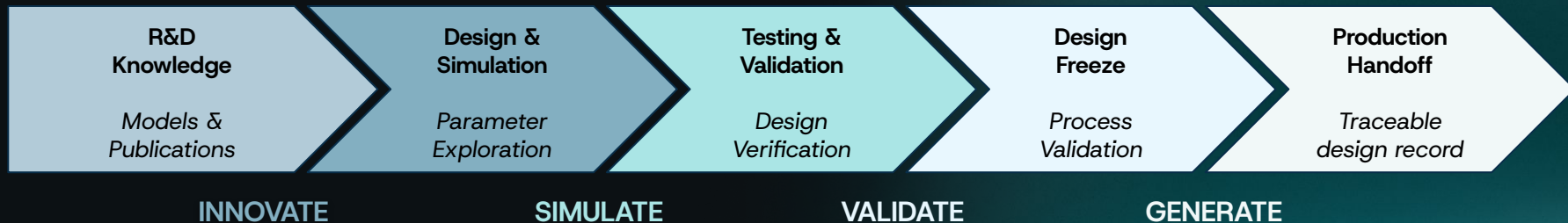
to model traceable industrial simulations & workflows

vs. wrangling multiple systems of record



End-to-End Development Traceability, with Protos.

Every design decision, simulation result, test data and parameter change – captured, versioned, and auditable across the **full product lifecycle**.



Version History

Every decision & change saved and labeled. Pull any parameter thread to any prior state instantly.

Data Traceability

Know exactly which model-parameter pair produced each result. Full attribution chain.

Interactive Auditing

One-click export of complete design history, decisions, and quality records. Bridging the Design-to-Process gaps

IDEAL CUSTOMER PROFILE

Protos

Your Co-Engineer for Industrial workflows

TARGET INDUSTRIES

Energy

Semiconductors

Robotics

Aerospace & Defence

Bio-pharma

COMPANY FIT & VALUE ALIGNMENTS

R&D-intensive

Engineers, physics-driven product performance

Slow iteration cycles

Days/weeks to simulate, expensive prototyping

Expertise bottleneck

Simulation locked to 1–2 specialists

Legacy tooling pain

COMSOL · Ansys · MATLAB — siloed, slow

Low Applicability of Generic LLM products

Requirement of structured data & decisions

CHAMPIONS & ECONOMIC BUYERS

Senior R&D Engineer

Electrochemical · Thermal

The simulation expert everyone waits on

Computational Engineer

FEA · CFD · Multiphysics

Builds pipelines. Models made traceable

Materials Scientist

Battery · Semicon · Bio

Designs stacks. Connects parameters

Process Engineer

Scale-up · Validation

Bridges R&D to production

Head of Product or Science R&D

Needs force multipliers. Disrupt.

Chief Technology Officer

The AI-native R&D bet.

Head of Engineering

Compresses design cycles.

Protos as Co-Engineer

"Built for engineers who work from physics."

- Runs physics-informed simulations autonomously
- Surfaces design risks before prototypes
- Distributes specialist knowledge
- Closes the design-to-production gap

Widely Applied Industrial Simulations

Electrochemical

Thermal Analysis

CFD

Multiphysics

FEA / Structural

Chemical Processes

Discrete Simulations

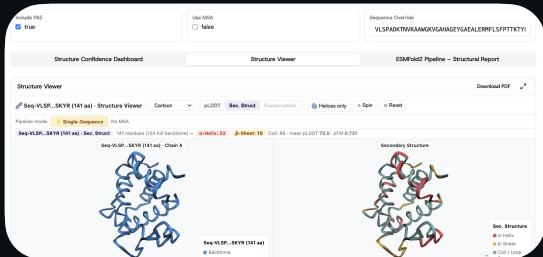
CAD Optimization

Operations Research

Route Optimization

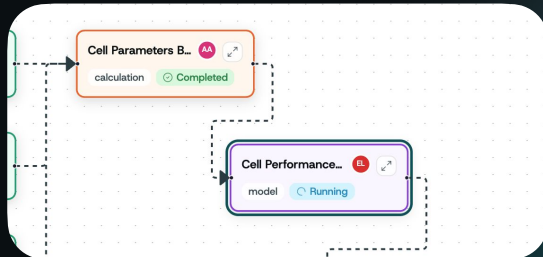
Supplier Selection

Protos Enterprise-grade Features



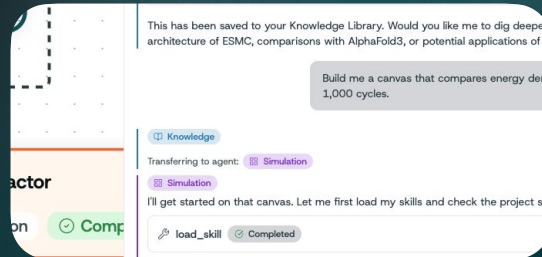
Publish & share results

Share full functioning workflows through a link



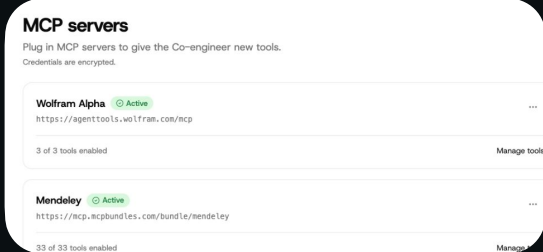
Real-time collaboration

Work with your team live on the same project



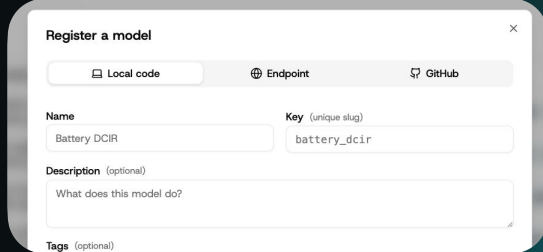
Specialist agents

Purpose-built agents with auto hand-offs



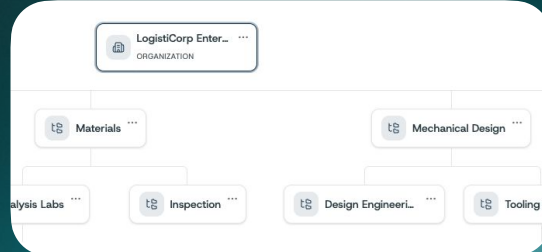
MCP-compatibility

Connect to any other tool through the Co-engineer



Own model integration

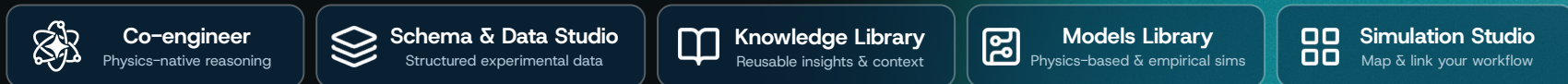
Register from code, GitHub or endpoint



Organizations and Role-based Access

Integrate your whole organization securely

THE FIVE CORE AGENTIC BUILDING BLOCKS FOR INDUSTRIAL R&D



Crossing Cross-Section Drawings – Harnessed Knee Joint Candidates

Geometric cross-sections per ISO 281 / ISO 76 conversions: Dimensions are representative (normalized scale)

Sealing Type	Rating Life Standard	F ₁₀ & S ₁₀ Values	Geometric Accuracy	Moment Load	Knee Suitability
Contact roller	ISO 281 / ISO 76	ISO 281 ISO 286	ISO 481	★★★★	Primary ✓
Wet Angular Contact	ISO 281 / ISO 76	ISO 286	ISO 482	★★★★	All ✓

[Explore here](#)

Swelling force at EOW vs structural limit

Condition	Swelling force (N)
T45	~45,000
T25	~20,000
T25+SOC	~19,000
T25+SOC+Rest	~18,000

Structural limit = 24,000 N

Explore here

[illegible]

Explore here

④ Phase Morphology Patterns

Canvas renderings: spinodal bicontinuous (HEA), vacancy-ordered checkerboard (YSZ), columnar dendrites (Ti-6Al-4V)

HEA	YSZ ZrO_2 -8%Y	Ti-6Al-4V
Spinodal bicontinuous	Vacancy-ordered checkerboard	Columnar dendrites

The figure displays three distinct phase morphology patterns. The first, HEA, shows a spinodal bicontinuous structure with a scale bar of 125 μm . The second, YSZ, shows a vacancy-ordered checkerboard pattern with a scale bar of 10 nm/tile. The third, Ti-6Al-4V, shows columnar dendrites with a scale bar of 500 μm domain. Below the renderings, the text indicates $\lambda_{\text{spin}} = 1.1 \text{ nm}$, BCC/FCC decomposition, Cubic sym. ($m=4$), Oxide sublattice, Melt pool solidification, and $T = 1923 \text{ K}$.

Explore here

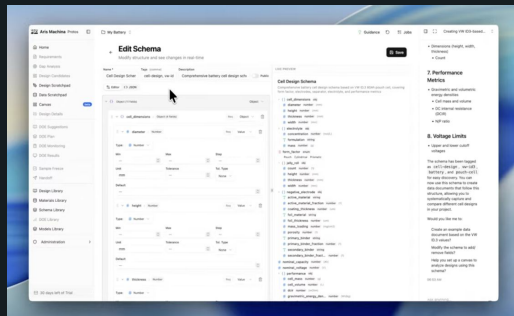
The screenshot shows a top-down view of a map in the Roblox game 'Squad'. The map is dark with various locations marked by icons and labels. On the left, there is a 'CLIENT INFORMATION' panel showing details for 'Player 1' (a female character) and 'Player 2' (a male character). Below this is a 'CLIENTS LEGEND' with icons for different types of clients (e.g., 'Client 1', 'Client 2', 'Client 3'). On the right, there is a 'CLIENTS LIST' panel showing a list of players with their names, avatars, and status (e.g., 'Player 1', 'Player 2', 'Player 3'). The map itself has several locations labeled, including 'Supply A', 'Supply B', 'Supply C', 'Supply D', 'Supply E', 'Supply F', 'Supply G', 'Supply H', 'Supply I', 'Supply J', 'Supply K', 'Supply L', 'Supply M', 'Supply N', 'Supply O', 'Supply P', 'Supply Q', 'Supply R', 'Supply S', 'Supply T', 'Supply U', 'Supply V', 'Supply W', 'Supply X', 'Supply Y', 'Supply Z'. There are also various icons for different types of vehicles and equipment scattered across the map.

Explore here

Automated Guided Vehicle routing on factory shop floor optimization study using Nvidia CUOpt model

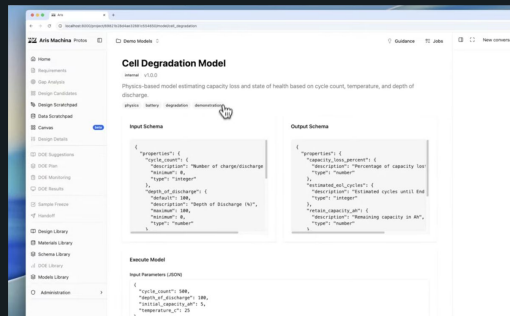
Enabling R&D engineers to utilize the power of Generative AI in day-to-day complex workflows

Manage all Master Data



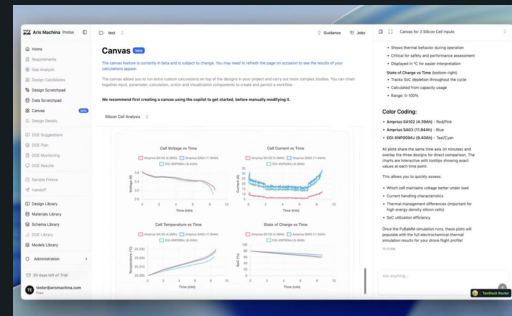
 [Schemas and Data studio](#)

Encode your Physics or AI



 [Custom Simulation Models](#)

Execute & Share Experiments



 [Simulation Pipelines](#)

Detailed Use-Cases Across Other Industries

 [BEES Cell Selection Pipeline](#)

 [Requirements Gap Analysis](#)

 [Bio-pharma / Cell Therapy Pipeline](#)

 [Drone Detection System](#)

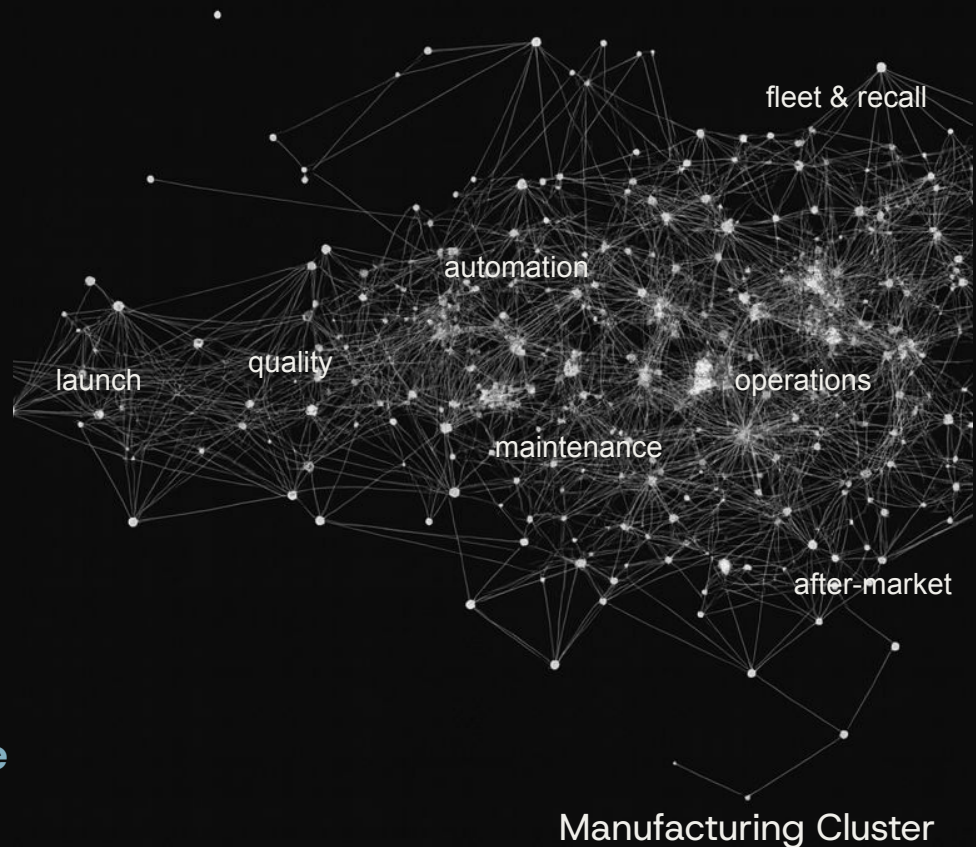
Advanced Factories struggle with mapping **Data to Knowledge**

Legacy software & data platforms
are stuck with incomplete context.



Gemba

AI-native Manufacturing Toolkit to maximize
operational efficiency





AI-native Shopfloor toolkit to maximize knowledge.

The Factory that never stops improving

Configurable Agents that Adapt

Every process set up with Gemba to monitor, learn, and correct. From quality drift to chronic downtime — caught before impact.

Factory Knowledge Graph without Blind Spots

Semantic ontology connects telemetry, images, logs, transactions, events and process results into one operational knowledge graph.

Non-disruptive by Design

Deploys alongside existing IT/OT systems (MES, SCADA) with open standards. Pilot ready in weeks, not quarters.

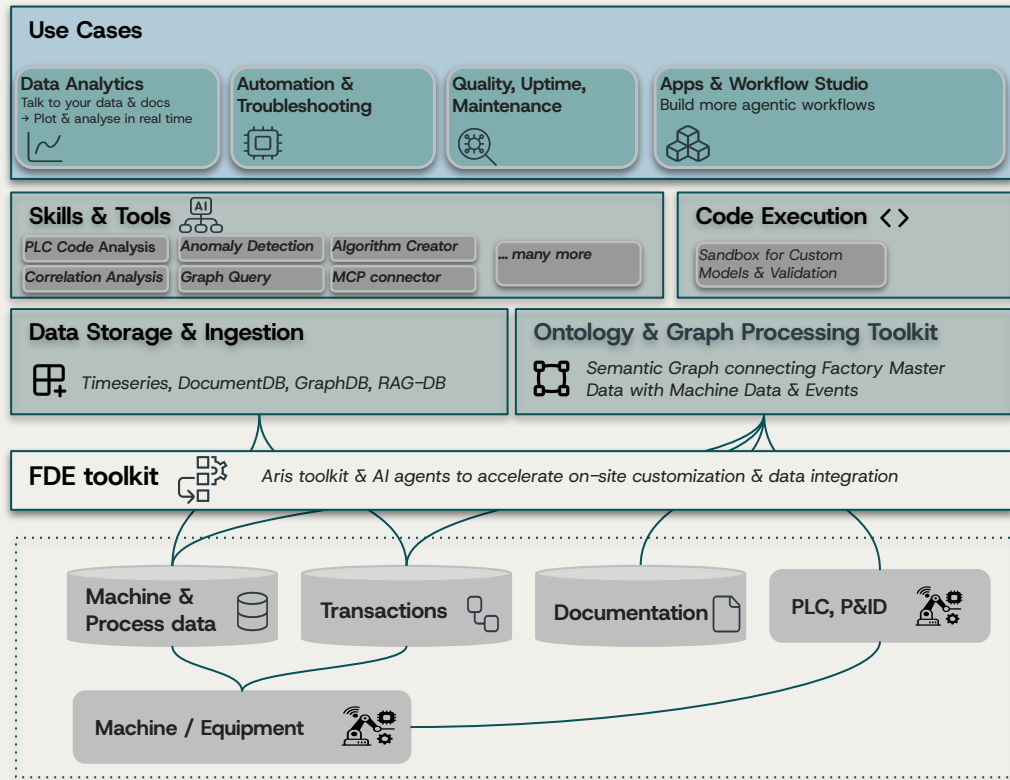
seconds

to data & root-cause for
any production event

vs.

days of manual investigation,
high scrap costs and downtime

The Gemba Architecture



 **Accurate** answers from complex data

 **Traceable** decisions rooted in factory data

 **Low latency** data architecture for bespoke AI workflows

 **Rapid setup** from diverse data-sources

 **Combine** Business, Machine Assets & Science in one Graph



Gemba Enables you to . . .

Configure Site-by-site

Site → Workcenter → Line → Module
or defined by the customer/user

Agentic Ingestion

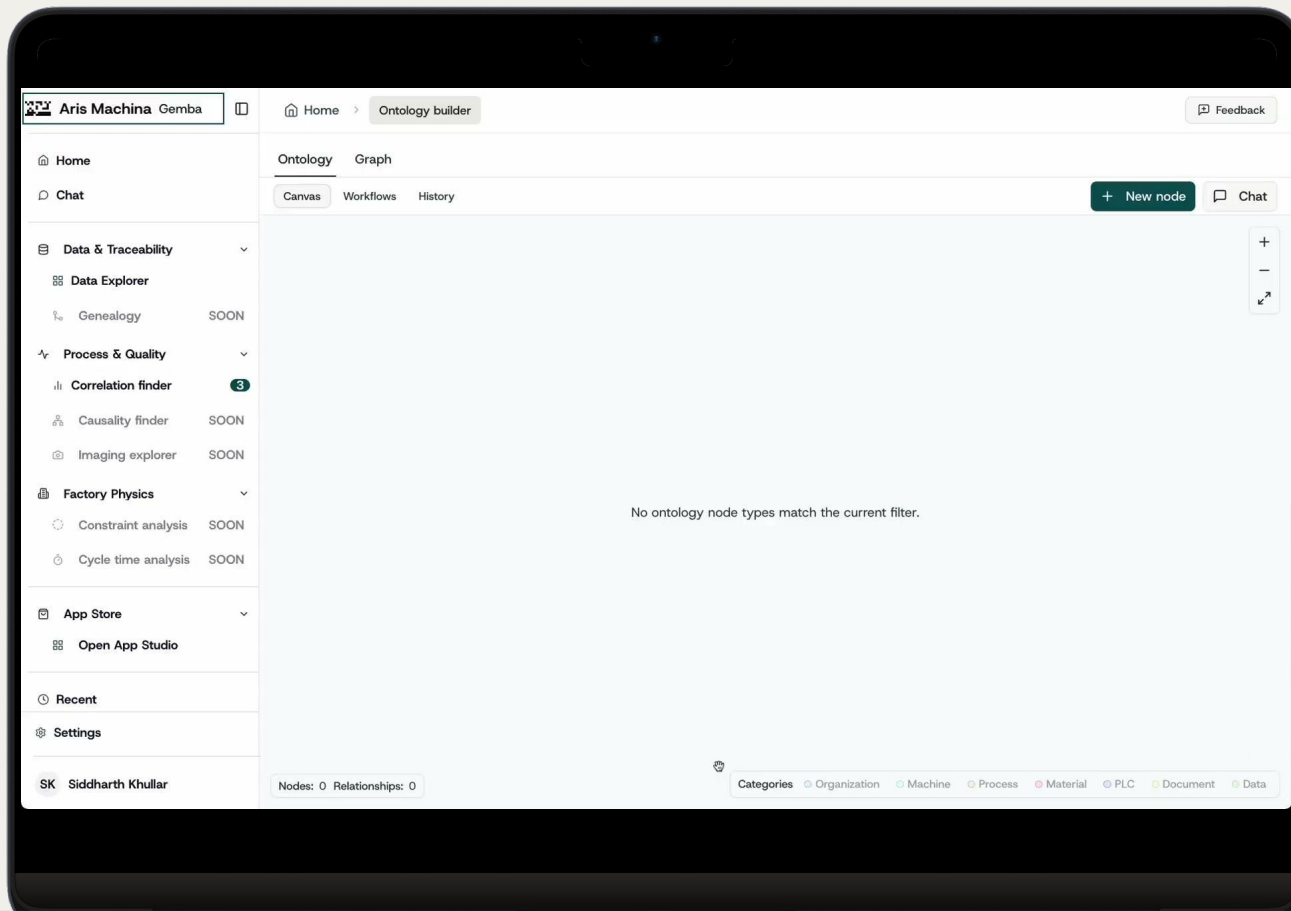
PLC → Equipment → Process →
Operations → Databases

Health Check & Monitoring

Verifying nodes, relationships, orphan
nodes in the relationship map /
ontology etc.

Change Management

A simple workflow to trace, track log,
and approve every change made to
the ontology



Aris Ontology: State of the Art (SOTA) vs. 3 ways we go beyond

Simple to adopt, built to scale — **manufacturing first, architected for the whole enterprise**

ALIGNED WITH SOTA

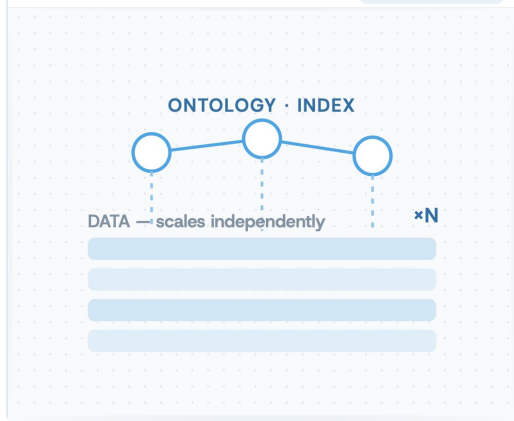
The shared baseline the whole field agrees on.



1 BEYOND SOTA

Graph Schema

1 index · N stores



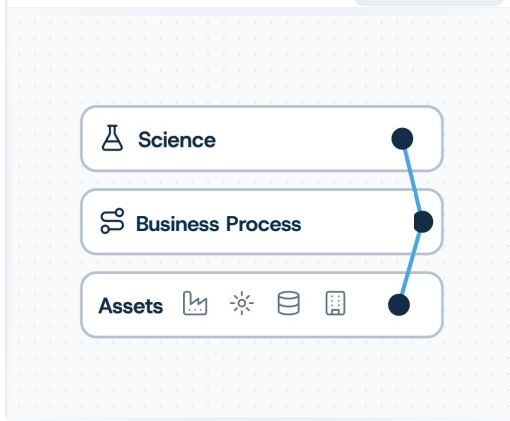
ONTOLOGY-AS-INDEX

Add data at any scale — no re-modeling to grow enterprise-wide.

2 BEYOND SOTA

Graph Layers

cross-domain



MULTI-LAYER BRIDGING

One answer across domains — because real problems are multifactorial.

3 BEYOND SOTA

Canvas Workflows History

1 uncommitted

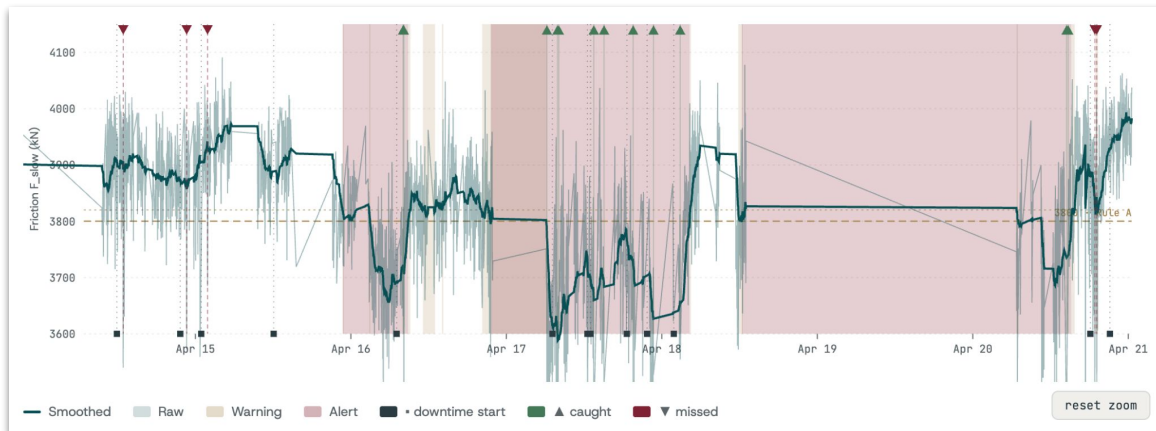
Commit

- **add alarms node to ontology**
hicham@arismachina.com · May
+1 node +2 edge +5 prop
- **Remove duplicated relationship**
anne-sophie@arismachina.com · May
-1 edge
- **Added new Graph Schema**
lukas@arismachina.com · May 7
+35 prop -7 prop

VERSIONED & AUDITABLE

Every change reviewed & reversible — audit-ready, not a prototype.

Model | Plunger Friction



How it works

Input data

- Shot payloads: Plunger displacement, Plunger velocity, Hydraulic & metal pressures.
- Past 200 shots for thresholding

The Physics

- Equation of motion
- Estimating friction to match shot curve

Output

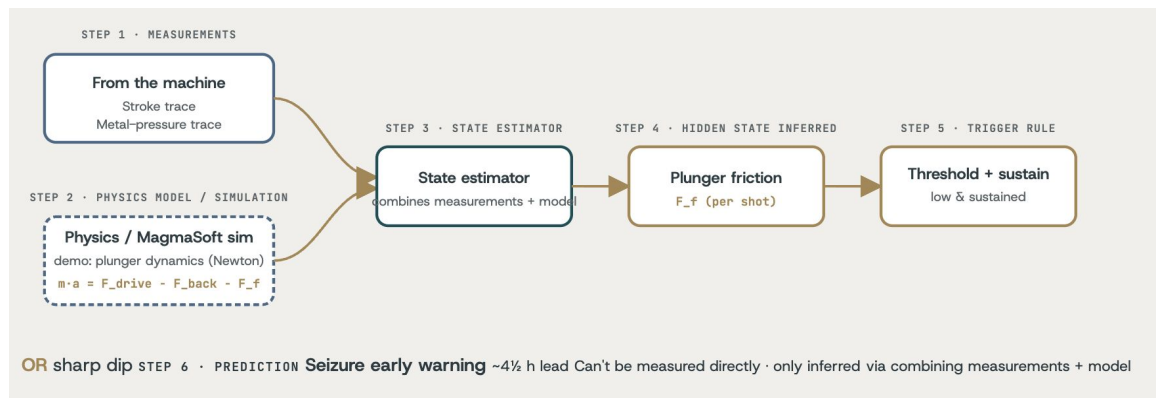
- Friction value for every shot
- Alert when friction deviates from baseline

How it is used

Business value – Friction warning predicts plunger seizure downtime -> **SHT:SZMON**

Delivery – Gemba warning workflow

Impact – 2% downtime predicted ~4.5 hrs lead time



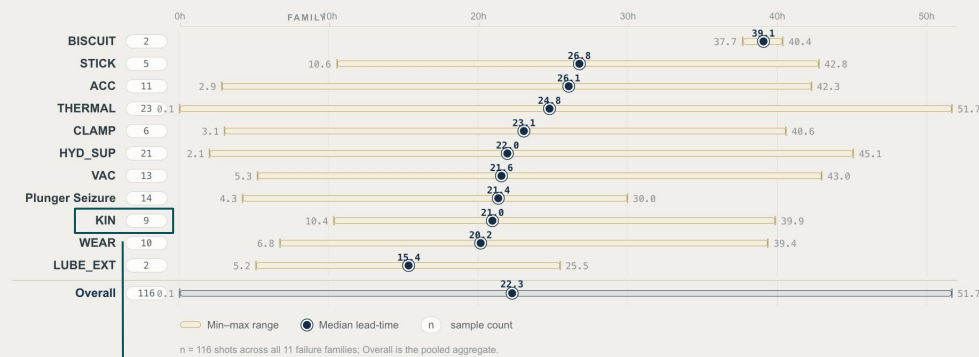
Model | Plunger Dynamics

How it works

SEIZURE EARLY-WARNING · SIGNAL FAMILIES

Lead-time by failure family

Warning-time distribution before failure, in hours — bar spans min → max, dot marks the median. Sorted by median.



Input data

- Shot payloads: Plunger displacement, Plunger velocity, Hydraulic & metal pressures.
- Process parameters

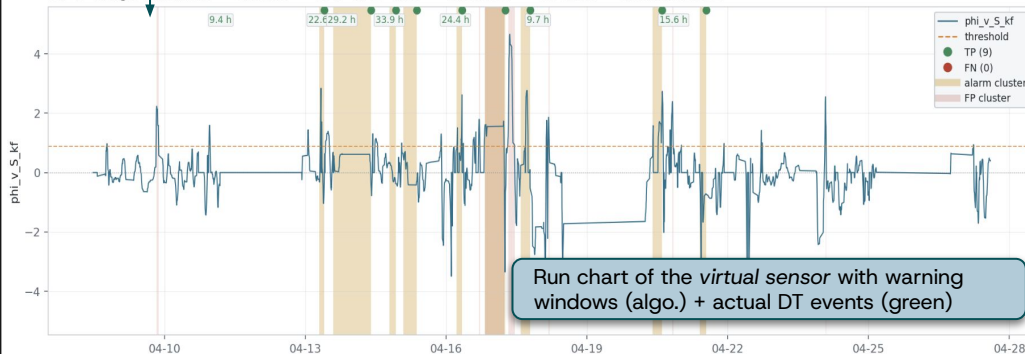
The Physics

- Equation of motion, wear kinetics, sleeve thermal, boundary lubrication

Output

- Virtual sensor values for plunger friction force, RMS acceleration, energy flow, metal pressure decomposition, multiplier health index, Stribeck velocity, plunger-tip clearance, polytropic-N₂ leak coefficient, thermal fouling resistance, and associated thresholds.

PFF-1 · Plunger kinematics · events 9 · TP 9 · FP 2 · FN 0 · clusters 17 · F1 0.900 · detector SINGLE



How it is used

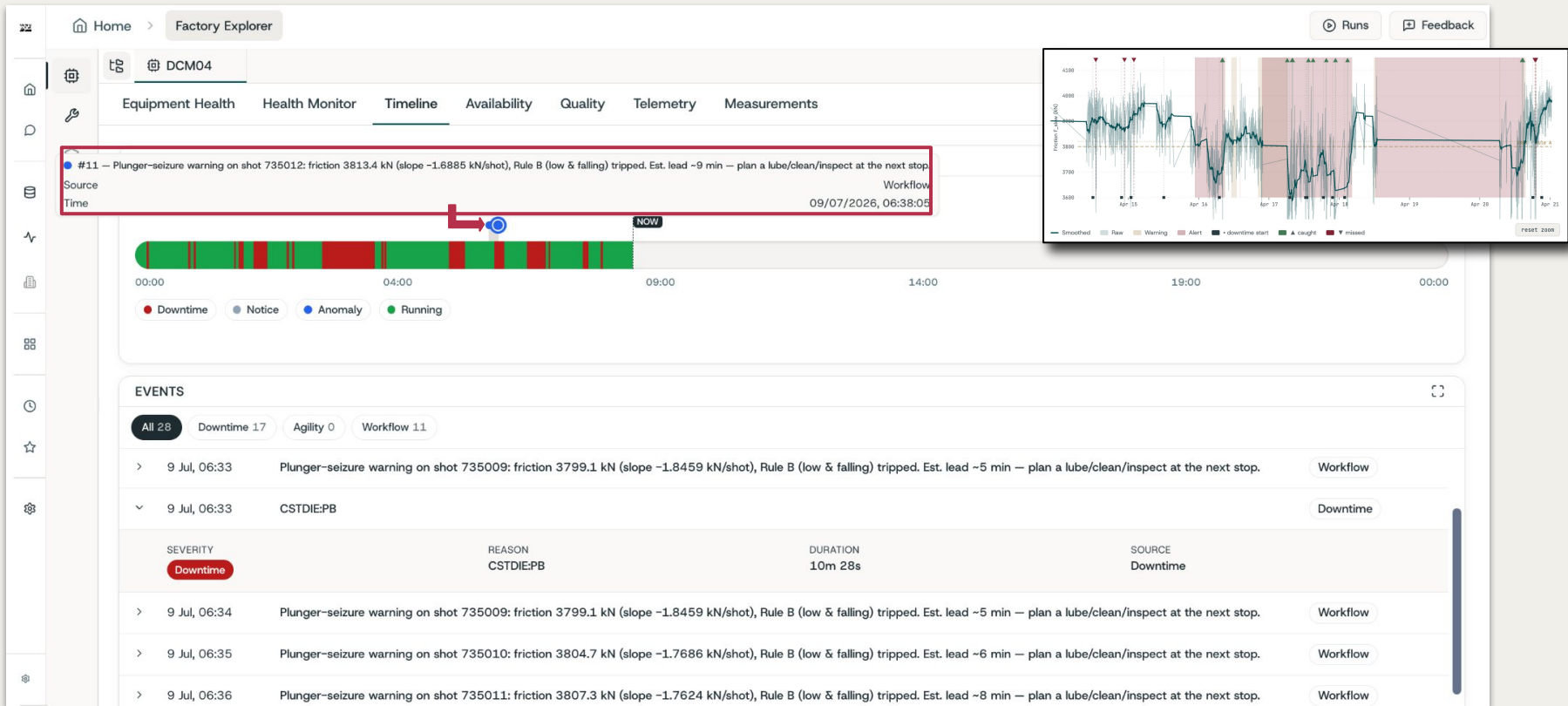
Business value: Virtual sensors correlates to downtime events → 20 downtime codes

Delivery: Gemba warning workflow

Impact: 9% downtime predicted ~22 hrs median lead time

Impressions from the week:

Gemba warning for *potential* plunger seizure issue (July 9 6:33am) – passively running in CCUK Gemba app



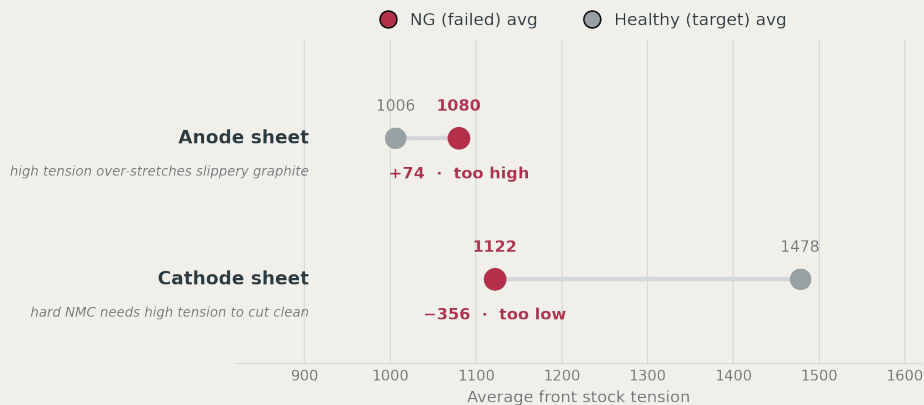
Use Case: Material and Cutter Tension

Gemba found one machine setting that fails two materials in opposite ways

PRODUCTION ISSUE

Anode and cathode sheets left the cutter with inconsistent tab widths, and no one knew why the two materials behaved differently. Gemba traced it to one setting: front stock tension.

WHAT GEMBA FOUND

**ANODE** *Stretchy***Tension ran too high****+74 units above target****CATHODE** *Stif***Tension ran too low****-356 units below target**

PROPOSED ACTIONS

- Set material-specific tension limits → raise cathode limit + lower anode limit
- Track tensions separately for both materials as a live SPC signals.
- Material change → triggers calibration per roll

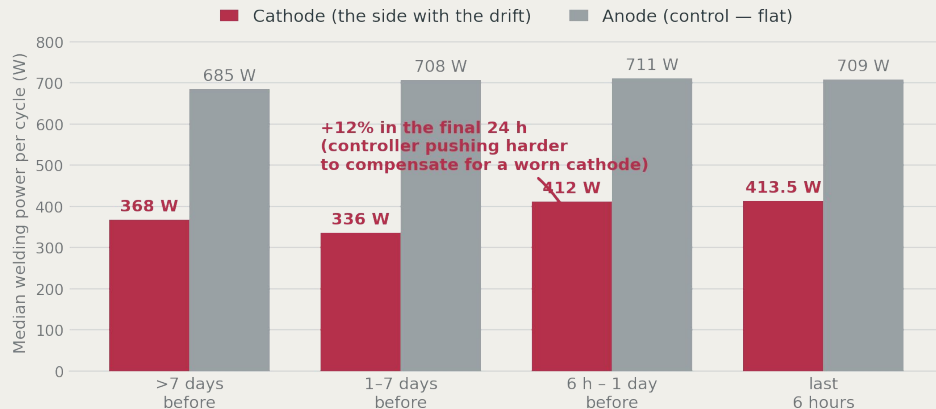
Use Case: Predictive Maintenance

Gemba found a second signal that a welding tip is worn

PRODUCTION ISSUE

Tips are swapped on a fixed cycle count. Gemba found the line's own power data also confirms real wear backing the scheduled swap with evidence, not just a counter.

WHAT GEMBA FOUND



WORN TIP (CATHODE)

Power climbed ~12%

In its final 24h – Anode stays flat

PROPOSED ACTIONS

Monitor both triggers: cycle counter + median welding power.

When power crosses the wear threshold, replace the tip then, even if the counter hasn't reached its scheduled swap point.

DOWNTIME SAVINGS

~109 hours
over 6 months

Reference Case

TARGET INDUSTRIES:

Energy

Semiconductors

Robotics

Aerospace & Defence

Bio-pharma

Context

INDUSTRY

Battery manufacturing

FACTORY

Cell assembly, 16 GWh/year

EQUIPMENT

High automation with OPC connection to MES for telemetry and quality.

PROCESS

Complex process with 10000+ process variables to root cause and solve quality, recall & downtime events

Challenges

Fragmented systems of production, engineering, and operations can't enable fast enough root causing without high operational impact.

Quality: Frequent recurring (chronic) quality issues resulting in 3-5% scrap of material, and occasional severe acute events requiring multi day/week investigation.

Downtime: Frequent chronic short duration downtime events without sufficient skilled operators and documentation, affecting OEE.

Aris Solutions

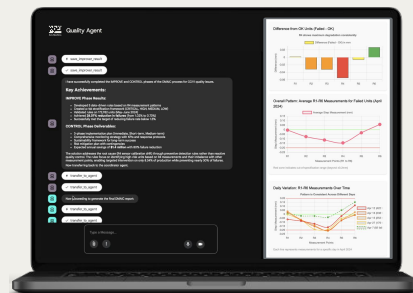
Gemba workspace unifies all production data streams into a single ontology

Gemba Quality: Agents continuously monitor the relationships between process variables, materials, and found relationships unseen in conventional process control methods.

Gemba Downtime: Tools were tested to operators to handle current events more robustly that normal machine manuals cannot perform. Proactive corrective actions are enacted to prevent downtime.

Quality Analysis Report Generation Multi-Agent System

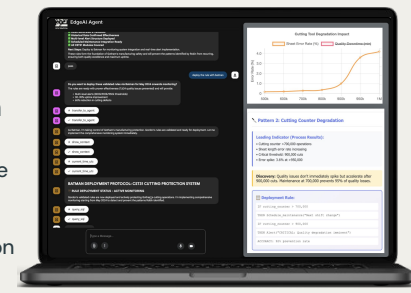
Diagnosing Root causes from telemetry, vision & alarms – autonomously



Maintenance Pattern and Rule Multi-Agent System

Pattern detection in large machine datasets.

Automatic rule generation and monitoring.



Tangible Customer Benefits

Protos

R&D & Design Intelligence



Quantitative

- Avoid Knowledge Fragmentation & Know-how Loss
- Reduced product development cycles
- Fewer design iterations
- Lower Testing Costs
- Decreased time-to-market

Qualitative

- Closed loop between design intent and manufacturing reality
- Physics-grounded modelling
- Preserved institutional knowledge
- Enhanced collaboration
- Faster innovation cycles

Gemba

Manufacturing Intelligence



Quantitative

- Increase Overall Equipment Effectiveness (OEE)
- Reduce scrap & Improved yield
- Rapid Root Cause Analysis
- Reduce downtime
- Lower integration costs

Qualitative

- Real-time operational transparency
- Shift from reactive to proactive troubleshooting
- Empowered factory teams
- Accelerated learning curves

**Combined
Benefits**

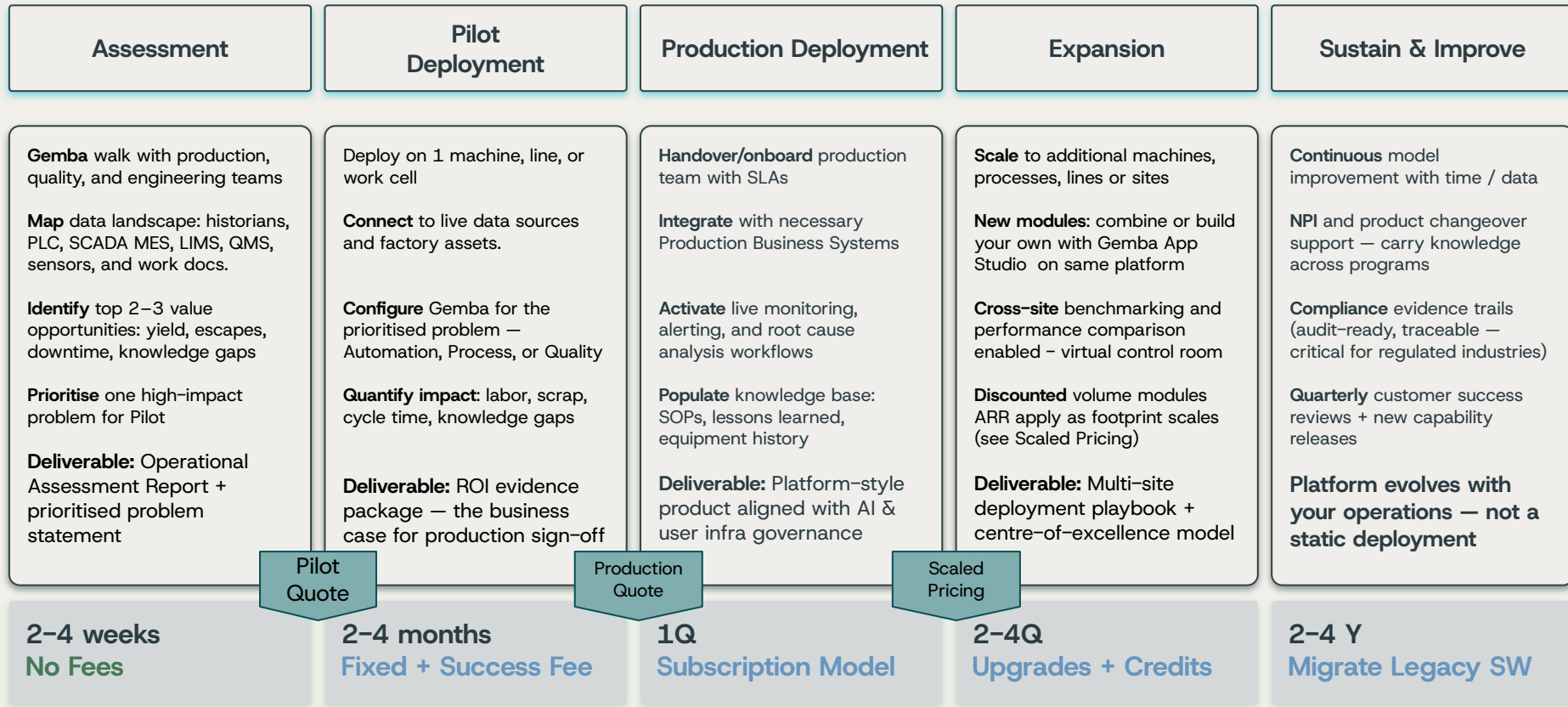


End-to-end product lifecycle intelligence
from concept through production

Compound benefits as design insights feed
production optimization and vice versa

Customer Journey

Aris aims to prove value early and scale according to your operational context and need



Gemba infrastructure scenarios

Three tiers to match any infrastructure, network constraint, or data governance requirement.

★ RECOMMENDED

	Option 1 Aris Private Cloud	Option 2 Customer Private Cloud	Option 3 Hybrid
Where Gemba runs	Aris-managed Cloud (GCP)	Fully on customer tenant (BYOC)	Core in Aris-managed cloud (GCP) + edge deployment on-site
OT Data (Machines)	Gemba connects OT layer via standard protocols to cloud	Customer Cloud	Gemba connectors deployed on-site near the OT infra
MES / IT Data (Events)	Data warehouse access or customer-managed replication	Customer Cloud	Data warehouse access or customer-managed replication
Data Governance	Aris managed	Customer Managed	Aris Managed
Setup Complexity	Low	Medium	Medium
Security	TLS in transit, GCP enterprise standards, Aris-managed tenant	Customer-controlled perimeter, network & InfoSec applies	TLS in transit, Gemba is outbound-only (no inbound ports)
Feasibility	Fastest to deploy; requires firewall rule or MQTT config	Moderate effort; customer-owned cloud / on-prem infra / compute	Moderate — Gemba requires on-site hardware install

Aris Competitive Differentiation

Unified ontology,
not fragmented tools

End-to-end
lifecycle coverage

Built by operators who've
scaled world-class factories

Foundation for frontier
technology factories

Agentic AI that acts,
not just analyzes

Avoids the "Industry 4.0
fatigue" data trap

A Paradigm Shift for Industry

- Analysis required **"just in time"** for action.
- ↳ **Unstructured data** utilized & valued
- ↳ Complex hidden **patterns** are uncovered
- ↳ Experts curate & **scale their knowledge**
- ↳ Shop floor is elevated & **up-skilled**
- ↳ **Ramp is enabled** with reduced effort & time

Ready when you are!

Contact ready@arismachina.com

Visit www.arismachina.com